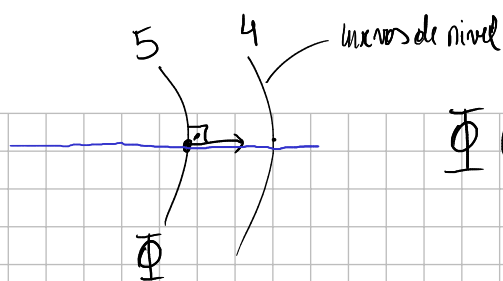
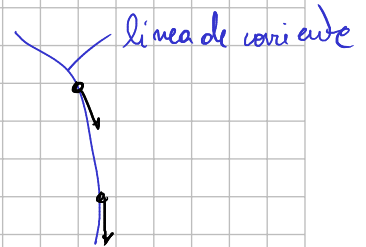


$$\underline{v} = \begin{bmatrix} u \\ v \end{bmatrix}$$



$$\Phi(x, y)$$

$$\Phi \text{ Función potencial} \left\{ \begin{array}{l} u = \frac{\partial \Phi}{\partial x}, \quad v = \frac{\partial \Phi}{\partial y}, \quad \underline{v} = \nabla \Phi \end{array} \right.$$



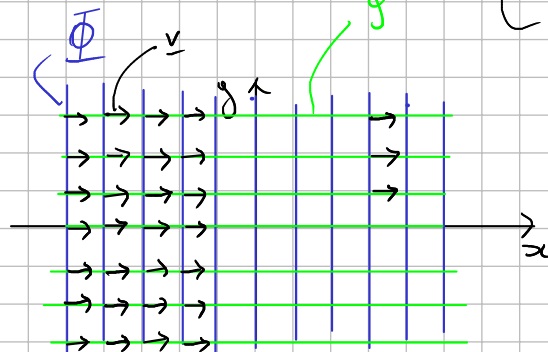
$$\Psi \text{ Función de línea de corriente} \left\{ \begin{array}{l} u = \frac{\partial \Psi}{\partial y} \\ v = -\frac{\partial \Psi}{\partial x} \end{array} \right.$$

$$\text{Función potencial } \Phi(x, y) \\ \Phi = x$$

$$\left\{ \begin{array}{l} u = \frac{\partial \Phi}{\partial x} = 1 \\ v = \frac{\partial \Phi}{\partial y} = 0 \end{array} \right.$$

$$\text{Función línea de corriente } \Psi(x, y) \\ \Psi = y$$

$$\left\{ \begin{array}{l} u = \frac{\partial \Psi}{\partial y} = 1 \\ v = -\frac{\partial \Psi}{\partial x} = 0 \end{array} \right.$$



$$\text{Pensemos que es un } \Phi(x, y) \text{ cuando} \\ \Omega_{ij} = \frac{1}{2} \left(\frac{\partial v_j}{\partial x_i} - \frac{\partial v_i}{\partial x_j} \right)$$

$$\begin{bmatrix} 0 & \Omega_{12} \\ -\Omega_{12} & 0 \end{bmatrix}$$

$$\Omega_{12} = \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$= \frac{1}{2} \left[\frac{\partial}{\partial x} \left(\frac{\partial \Phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \Phi}{\partial x} \right) \right] = 0 \Rightarrow \text{Flujo irrotacional}$$

No es necesario
más tiempo
con

$$\text{usamos } \Psi(x, y)$$

$$\Omega_{ij} = \frac{1}{2} \left[\frac{\partial}{\partial x} \left(-\frac{\partial \Psi}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \Psi}{\partial y} \right) \right] = -\frac{1}{2} \left[\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} \right]$$

$$\text{en particular, en este caso, } \Omega_{ij} = -\frac{1}{2} \left[\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} \right] = 0 \Rightarrow \text{Flujo irrotacional}$$

$$V_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)$$

$$V_{11} = \frac{\partial u}{\partial x}$$

$$V_{22} = \frac{\partial v}{\partial y}$$

$$V_{12} = V_{21} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)$$

an der Wand es

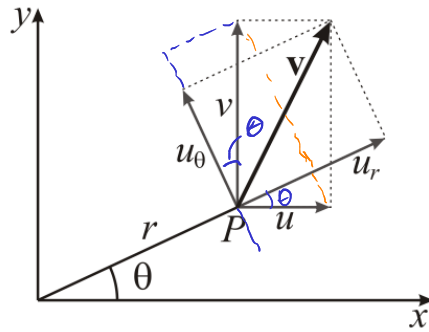
$$V_{11} = 0$$

,

$$V_{22} = 0$$

,

$$V_{12} = 0$$



$$u_\theta = v \cdot \cos \theta - u \cdot \sin \theta$$

$$u_r = v \cdot \sin \theta + u \cdot \cos \theta$$

$$x = r \cdot \cos \theta$$

$$y = r \cdot \sin \theta$$

$$\Phi(r, \theta) = \Phi(x(r, \theta), y(r, \theta))$$

$$\frac{\partial \Phi(r, \theta)}{\partial r} = \frac{\partial \Phi(x, y)}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial \Phi(x, y)}{\partial y} \cdot \frac{\partial y}{\partial r} =$$

$$= u \cdot \cos \theta + v \cdot \sin \theta = u_r \Rightarrow u_r = \frac{\partial \Phi(r, \theta)}{\partial r}$$

$$\frac{\partial \Phi(r, \theta)}{\partial \theta} = \frac{\partial \Phi(x, y)}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial \Phi(x, y)}{\partial y} \cdot \frac{\partial y}{\partial \theta} =$$

$$= -u \cdot r \sin \theta + v \cdot r \cos \theta = r \left(v \cdot \cos \theta - u \cdot \sin \theta \right)$$

$$= r \cdot u_\theta \Rightarrow$$

$$u_\theta = \frac{1}{r} \cdot \frac{\partial \Phi(r, \theta)}{\partial \theta}$$

Similar procedimiento para obtener los componentes de la velocidad en coordenadas polares en función de x y y .

$$u_\theta = - \frac{\partial \psi(r, \theta)}{\partial r}$$

$$u_r = \frac{1}{r} \frac{\partial \psi(r, \theta)}{\partial \theta}$$